11 (a) Fig. 11.1 shows a solenoid connected to a battery.

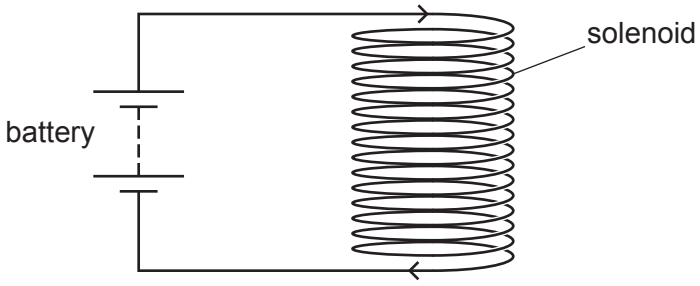


Fig. 11.1

On Fig. 11.1, draw the pattern of the magnetic field inside and around the solenoid. Indicate the direction of the magnetic field with an arrow. [3]

(b) Electrical power is transmitted at a voltage of 400 kV. A transformer reduces the voltage to 33 kV for use by heavy industry in large factories. The number of turns on the primary coil of the transformer is 11 000.

Calculate the number of turns on the secondary coil of the transformer.

number of turns = ... [2]

[Total: 5]

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